CS 391
Web Application Development

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Week - 9

Outline

1. Bootstrap with ReactJS
2. Reactstrap

Bootstrap with ReactJS

1. Bootstrap with ReactJS
2. Reactstrap

How to use Bootstrap with ReactJS?

- There are the following ways you can use Bootstrap in the ReactJS Project:
  1. Using Bootstrap CDN
  2. Install Bootstrap as a dependency
     - https://github.com/twbs/bootstrap : *npm install bootstrap*
  3. Use a package that includes rebuilt Bootstrap components that can be used as React components. The following are the two most common packages:
     - *react-bootstrap* : https://react-bootstrap.github.io/
     - *reactstrap* : https://reactstrap.github.io/

Bootstrap Usage with JavaScript frameworks

- While the Bootstrap CSS can be used with any framework, the *Bootstrap JavaScript* ***is not fully compatible*** with JavaScript frameworks like React, Vue, and Angular which assume full knowledge of the DOM.
- Both Bootstrap and the framework may attempt to mutate the same DOM element, resulting in bugs like dropdowns that are stuck in the "open" position.

Bootstrap Usage with JavaScript frameworks

- To best use ReactJS, it would be ideal to import Bootstrap classes as React components.
- Luckily, this is done by utilizing third-party libraries like *react-bootstrap* and *reactstrap,* to do so.

https://www.creative-tim.com/blog/web-design/how-to-use-bootstrap-with-reactjs/

Bootstrap libraries…

- Reactstrap and React-bootstrap, provide us Bootstrap Components without the need to use jQuery or JavaScript with direct DOM manipulation.
  - https://react-bootstrap.github.io/
  - https://reactstrap.github.io/

reactstrap

…Bootstrap libraries

- Reactstrap makes use of class components whereas React-bootstrap makes use of functions and hooks.
- The user may choose either depending on their preferences.

React-bootstrap vs Reactstrap

- Useful Links
  - https://drasticcode.com/difference-between-reactstrap-and-react-bootstrap/
  - https://dev.to/cassiolacerda/using-bootstrap-in-reactjs-reactstrap-or-react-bootstrap-gng
  - https://npmcompare.com/compare/bootstrap,react-bootstrap,reactstrap
  - https://www.creative-tim.com/blog/web-design/how-to-use-bootstrap-with-reactjs/
  - https://www.pluralsight.com/guides/how-to-integrate-bootstrap-with-reactjs
  - https://www.pluralsight.com/guides/how-to-use-bootstrap-components-with-reactjs
  - https://www.digitalocean.com/community/tutorials/react-fancy-forms-reactstrap#getting-started
  - https://www.copycat.dev/blog/reactjs-bootstrap/

Main Concept…

- **React-bootstrap** and **Reactstrap** are looks similar and both gives the well known "bootstrap" features.

Bootstrap Using HTML styles/classes

```
<div class="container">
  <div class="row">
    <div class="col-sm">
      Col 1 of 3
    </div>
    <div class="col-sm">
      Col 2 of 3
    </div>
    <div class="col-sm">
      Col 3 of 3
    </div>
  </div>
</div>
```

Bootstrap Using React-based Components

```
<Grid>
  <Row>
      <Col sm>
        Col 1 of 3
    </Col>
    <Col sm>
        Col 2 of 3
    </Col>
     <Col sm>
        Col 3 of 3
    </Col>
  </Row>
</Grid>
```

Reactstrap

1. Bootstrap with ReactJS
2. Reactstrap

Install Reactstrap

- Installation:
  - npm install reactstrap
- Reactstrap currently requires React 16.8 or higher.
- Reactstrap is currently compatible with Bootstrap 5.1.
  - If you are using Bootstrap 4, you'll need to use Reactstrap v8

https://reactstrap.com/
https://reactstrap.github.io/

Include Bootstrap

- Import Bootstrap in your application code:
  - npm install --save bootstrap
- Open your main js file (layout.js or page.js), add following import statement. It enables bootstrap styling
  - import 'bootstrap/dist/css/bootstrap.min.css’;

Reactstrap Components

- Accordion
- Alert
- Badge
- Breadcrumb
- Button
- Card
- Carousel
- Collapse
- Dropdown
- Fade
- Forms
- InputGroup
- Layout
- List
- ListGroup
- Modal
- Nav
- Navbar
- Offcanvas
- Pagination
- Popover
- Progress
- Spinner
- Table
- Toast
- Tooltip

Import Components

```
import React from 'react';
import { Button } from 'reactstrap';

export default (props) => {
  return (
    <Button color="danger">Danger!</Button>
  );
};
```

Example…

```
import React from 'react';
import { Container, Row, Col } from 'reactstrap';
import ProductCard from './ProductCard';
const ProductList = (props) => {
    const { data } = props;
    return (
        <Container>
            <Row>
                {data.map((item, index) => {
                    return (
                        <Col key={`product-${index}`} sm="6" md="4" lg="2" >
                            <ProductCard product={item}></ProductCard>
                        </Col>);
                })}
            </Row>
        </Container>
    );
};
export default ProductList;
```

...Example

```
import React from 'react';
import { Card, CardImg, CardText, CardBody, CardTitle, Button } from 'reactstrap';

const ProductCard = (props) => {
  const { product } = props;
  const baseUrl = '/images/';
  return (
    <div>
      <Card>
        <CardImg src={`${baseUrl}/${product.img}.jpg`} />
        <CardBody>
          <CardTitle tag="h5">{product.name}</CardTitle>
          <CardText>{product.description}</CardText>
          <Button color="primary">Order</Button>
          <Button color="danger">Delete</Button>
        </CardBody>
      </Card>
    </div>
  );
};
export default ProductCard;
```

Any Question?

Thank you for your attention

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